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Oil & Gas Fundamentals

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Oil & Gas Fundamentals

This course provides participants with a comprehensive introduction to oil and gas business of a company. Participants will learn the oil and gas fundamentals, understand oil and gas company’s operations and how various factors contribute to its performance. The program covers technical, economic and commercial aspects of oil and gas business, essential for any employee of an oil and gas company to understand. The course includes practical case studies and testing.

LOCATION

CLASSROOM & VIRTUAL DELIVERY

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Introduction

- Overview of upstream, midstream, downstream, and petrochemicals
- International and national petroleum companies
- Integrated and independent petroleum companies
- Oilfield services and subcontractors
- Crude oil and natural gas markets

–Consumption, production, reserves
–OPEC and other producing countries

Petroleum geology

- Geologic time
- Plate tectonics
- Rock cycles, rock types
- Formation of oil and formation of gas
- Hydrocarbon system and its elements
- Hydrocarbon reservoirs: types, quality (porosity and permeability)
- Conventional and unconventional reservoirs

Exploration

- Exploration process: onshore, offshore
- Success rates
- Exploration program: onshore, offshore
- Geophysics: G&G studies, 2D, 3D and 4D seismics
- Drilling: rig types, process, mud systems, casing and tubing, pressure control
- Exploration wells and data analysis
- Appraisal wells and data analysis
- Evaluation of a discovery: technical and economic
- Exploration outcomes: commercial discovery, relinquishment, retention

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Development and production

- Field development plan and FEED: onshore, offshore
- Planning and procurement
- A typical field's architecture and operations: oil fields, gas fields, onshore and offshore (shallow water, deep waters)
- Production facilities

–Producing and injection wells, gathering systems, separation systems, storage, metering, energy systems

- Operational

–Operating expenditure categories, leasing, general and administrative expenditure

- Decline rates, forecast of production
- Recovery rates and phases of field's production: primary, secondary and tertiary
- Reservoir engineering
- Artificial lift
- Well workover and stimulation
- Tertiary recovery methods
- Decommissioning and abandonment: onshore, offshore
- Tie-in development
- Shared infrastructure

Crude oil marketing and refining

- Lifting of production: overview and procedures
- Crude oil transportation: pipeline, modular (fleet type and sizes)
- Storage
- Crude oil sales

–Crude oil quality, prices and markets

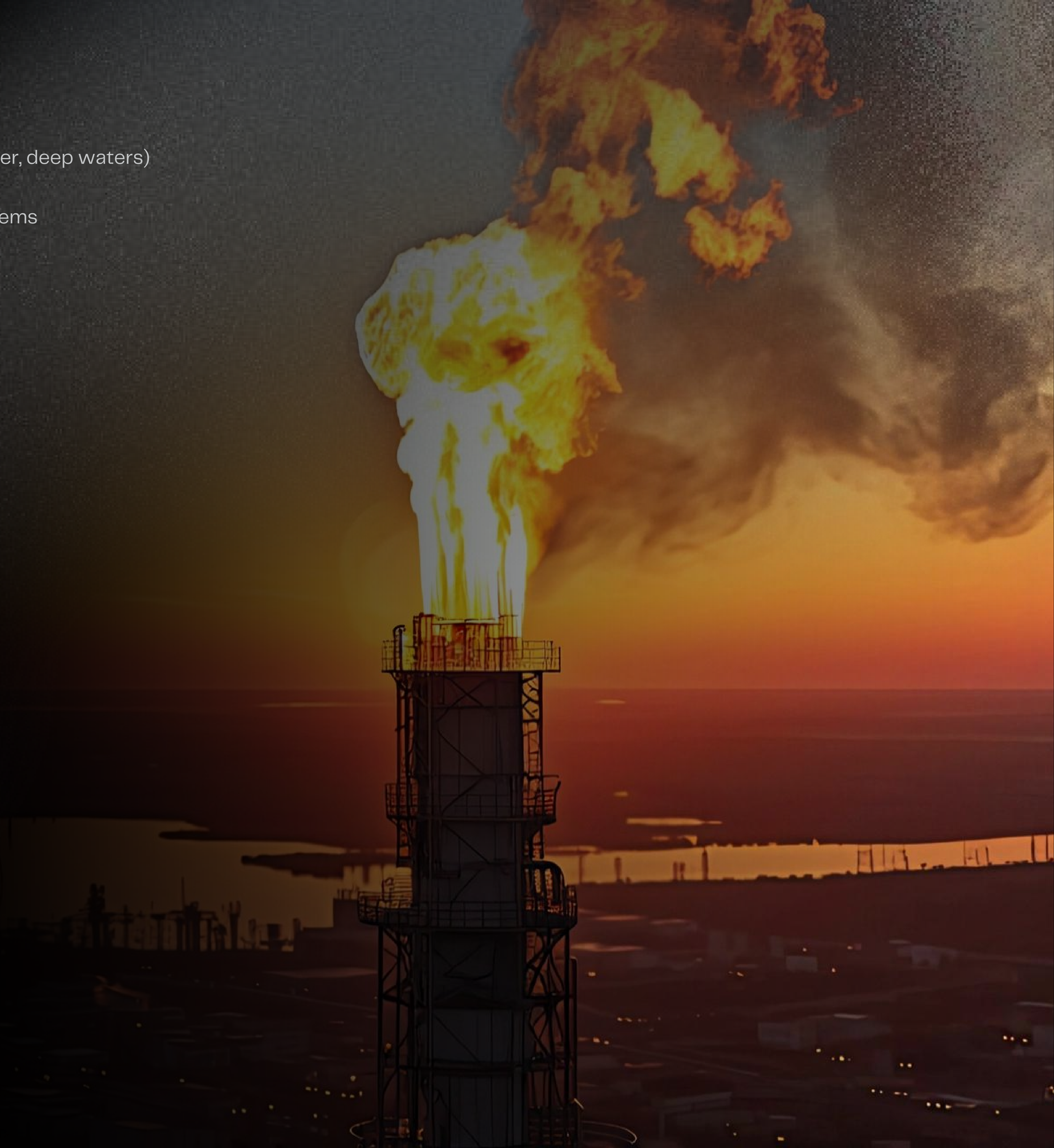
–Domestic sales and exports

–Principles and commercial structures

- Crude oil refining

–Processes and petroleum products

–Types of petroleum products, uses and markets



Natural gas marketing

- Quality of natural gas

–Composition of natural gas (impurities, liquids, helium)
–Associated natural gas from oil wells
–Gas plants and processing process
§ Methane: uses, markets and prices
§ Natural gas liquids: products, uses, markets and prices
§Natural gas storage
–Short-term storage
–Long-term storage

- Natural gas sales
  - Pipeline: overview and commercial framework, case studies
  - Liquified natural gas

–Value chain of an LNG project: upstream, liquification, transportation, regasification
–Liquification: offshore and onshore, integrated and hubs, scales, storage
–Transportation: fleet and sizes
–Markets, pricing and costs
–Contractual framework
–Case studies

Petrochemicals

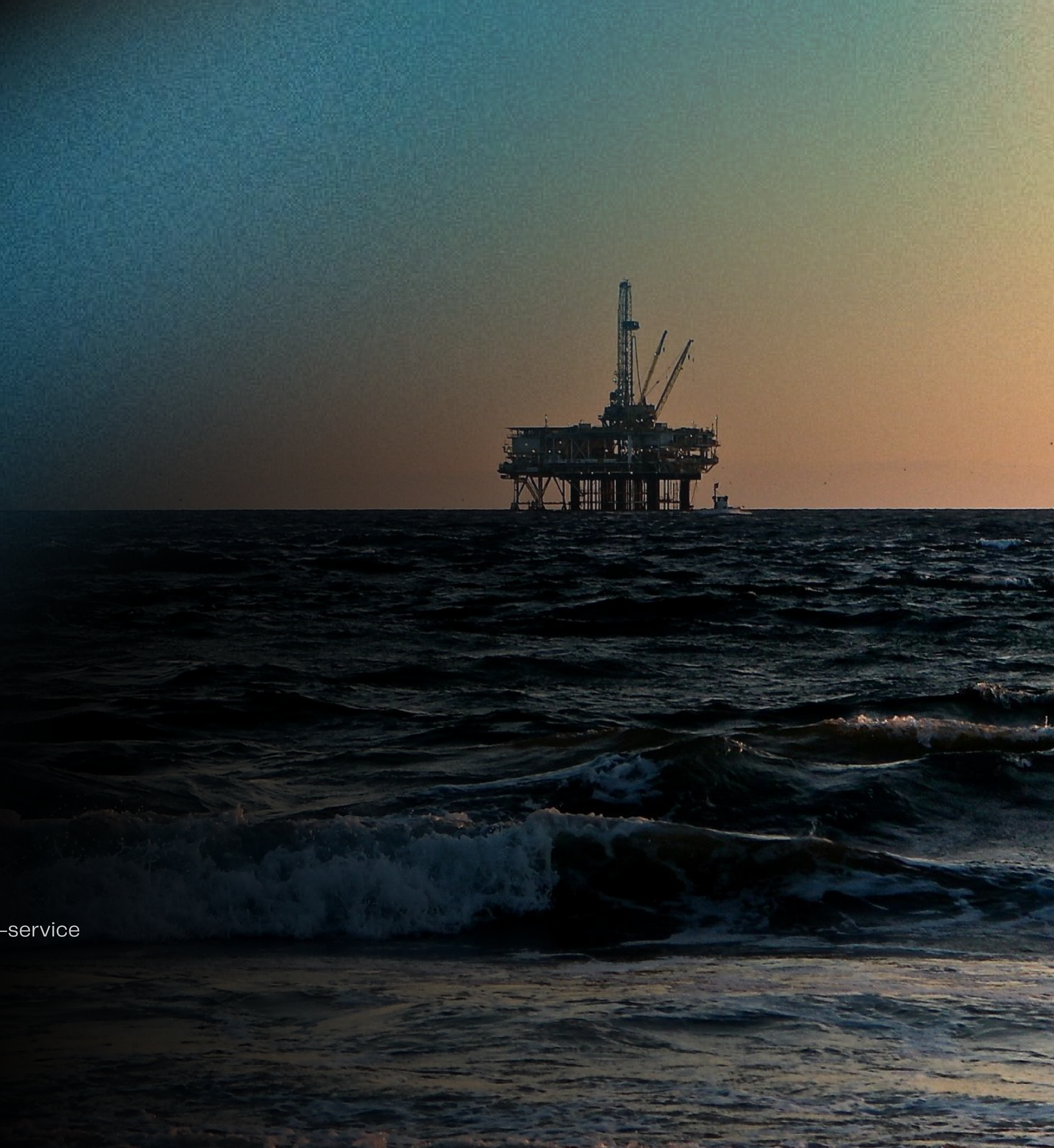
§ Feedstock
§ Primary petrochemicals
§ Intermediaries, derivatives and end-products

Petroleum contracts and regulatory framework

- Government's regulations of oil and gas operations: regulatory, technical, economic
- Allocation of oil and gas rights: how petroleum acreage is allocated to investors
- Overlapping rights: unitisation and joint development
- Petroleum contracts used by governments: concessions, production sharing contracts, risk-service contracts

–Understanding mechanics, differences and similarities
–Impact of each contract's type on investors

- Important provisions of petroleum contracts
- Permits



Financial and fiscal terms under petroleum contracts

- Introduction

–Nature and scope of upstream fiscal payments

–Project's value and fiscal terms

- Bonuses, acreage fees, training and other payments
- Royalties (rates, differentiation of rates, calculation rules)
- Taxes

–Understanding key calculations rules

- Production sharing

–Understanding the mechanics of production sharing, annual work programme cycles and approval processes

–Understanding cost recovery

–Understanding profit production sharing

- Special resource payments (windfall, additional taxes, R-factors, IRR-based calculations)
- Payments on subcontractors
- Domestic market obligations
- Local content obligations
- Service fees
- Environmental fees, methane emissions fees and carbon taxes
- Impact of such payments on company's operations
- Case study: Kurdistan, Federal Iraq, Abu Dhabi

Finance and economics

- Company's financials and shareholder's value
- Explanation of financial and economic indicators used by investors to evaluate oil and gas projects (NPV and discount rate, WACC and IRR, EMV, maximum exposure, payout, breakeven price, netbacks, others)
- Other metrics: daily and cumulative production, equivalent production, entitlement production
- Reserves

–Resources and reserves

–Classification of reserves

–Booking of reserves, revisions, importance for oil and gas companies

–Reserves and oil and gas prices

- Upstream performance measures
- Upstream cost structure
- Financing of upstream projects: exploration, post-exploration (reserve-based financing, farmouts, other)
- Financing of downstream
- Marginal oil and gas projects

Project profiles

- Conventional oil and gas: onshore, offshore
- Marginal projects: various types based on quality of geology and logistical environments
- Heavy oils
- Shale and low-permeable reservoirs
- Late-life and incremental production projects

Joint ventures

- Why companies have joint ventures
- Types of joint ventures
- Operators and non-operators
- Mechanics of joint ventures
- Farm-in and farm-outs
- Carried interest
- Governments, national petroleum companies and joint ventures

Risk management

- Major risks in upstream
- Major risks in downstream
- Risk mitigation strategies

Energy transition and impact on oil and gas businesses

- Key developments and impact on oil and gas business
- Greenhouse gas (GHG) emission

–Scope 1, 2 and 3 of GHG emissions for upstream and downstream
–Methane emissions and gas flaring
–LNG and emissions
–Emission intensity of operations (upstream, downstream)
–Case studies

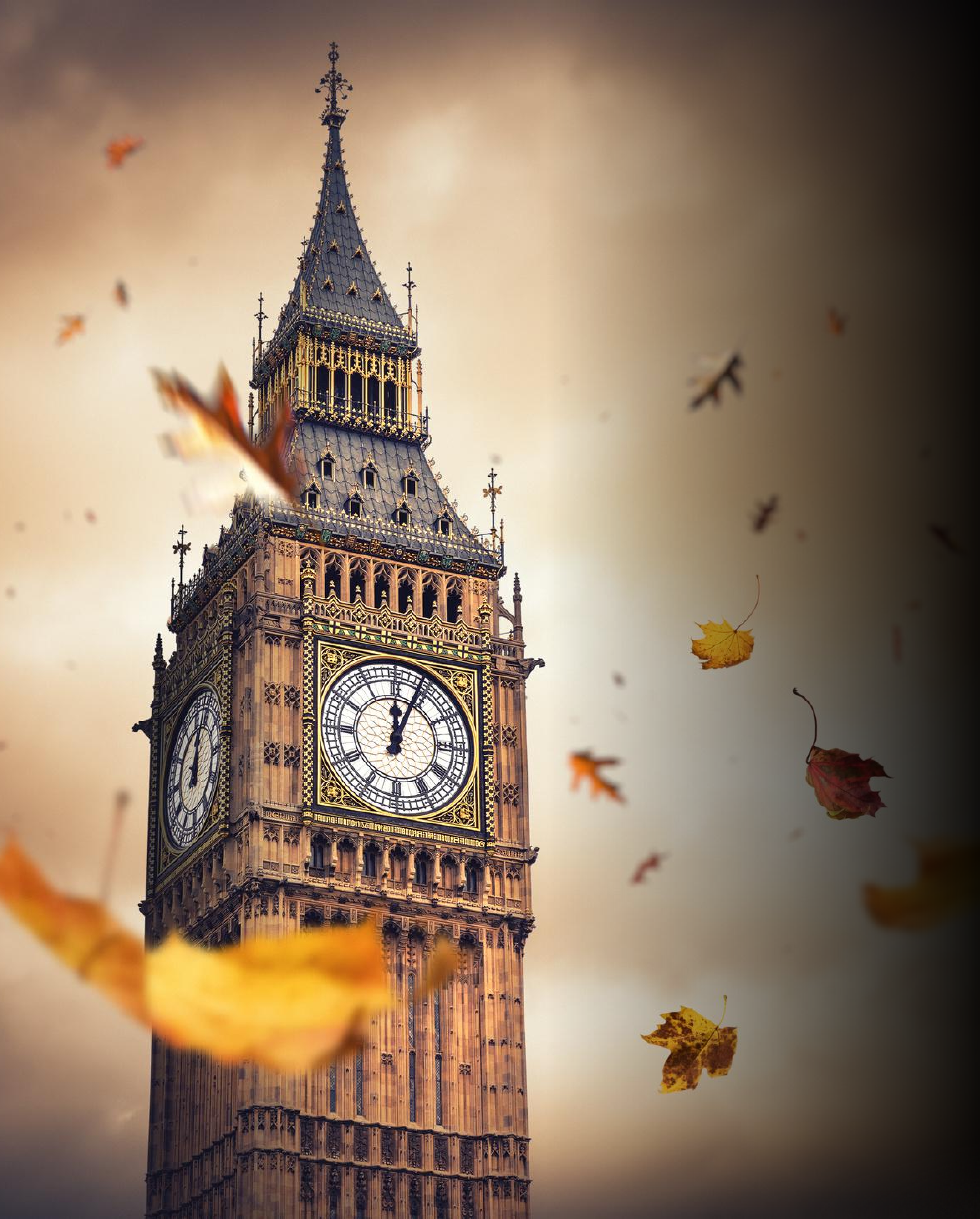
- Carbon pricing (incl. carbon border adjustments) and low-carbon regulations
- Carbon capture, utilisation and storage (CCUS)

–Overview of CCUS: value chain (integrated projects, storage hubs, service models), monetisation options, markets and costs
–Natural gas and CCS
–Oil and gas fields and infrastructure and CCS

- Low-carbon businesses of oil and gas companies

–Renewable energy: standalone, upstream
–Hydrogen (green, geologic)
–Lithium, other
–CCUS
–Case studies

- Energy matrix, outlook for demand and markets for oil and gas and derivatives



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